

Reproduction & Heredity

Why do you have your mom's eyes or your grandpa's smile? The answer is hidden in tiny instructions passed from parents to offspring. Let's crack the code of how traits are inherited — and how living things make more of themselves.

 Georgia Standard S7L3

WHAT YOU'LL LEARN

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| 01 Traits and Heredity | 05 Asexual Reproduction |
| 02 Genes and Chromosomes | 06 Sexual Reproduction |
| 03 Dominant and Recessive | 07 Selective Breeding |
| 04 Punnett Squares | 08 Putting It All Together |

3.1 Traits and Heredity

A **trait** is a feature you can observe — eye color, hair type, height, or the shape of an earlobe. **Heredity** is the passing of traits from parents to their offspring. It's the reason kids often look like their parents.

You inherited a set of instructions from each of your parents. These instructions decide many of your traits. Some traits, like eye color, come almost entirely from your genes. Others, like how tall you grow, come from both your genes **and** your environment (such as the food you eat).

! Did You Know?

You share about 99.9% of your DNA with every other human on Earth. That tiny 0.1% difference is what makes each person unique — different faces, voices, and traits.

✗ Common Mix-Up

Not all traits are inherited. A scar, a sunburn, or a pierced ear are **acquired** during your life — they aren't in your genes, so you can't pass them to your children.

3.2 Genes and Chromosomes

Your traits are written in a chemical code called **DNA**. DNA is organized into structures that fit inside the nucleus of every cell, like a set of nesting dolls.

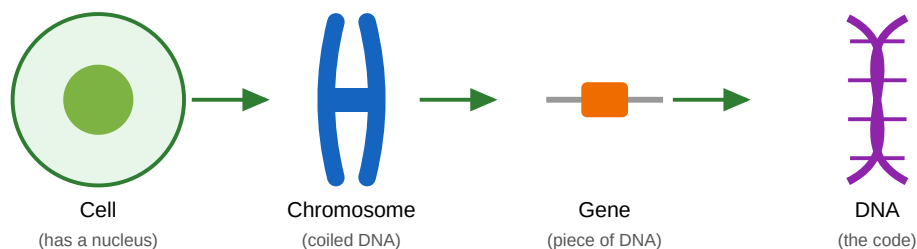


Figure 3.1 — From cell to DNA. Each structure fits inside the one before it.

Term	What It Is
DNA	The chemical code that carries all the instructions for life.
Gene	A short section of DNA that controls one trait, like eye color.
Chromosome	A long, tightly coiled strand of DNA. Genes sit along it like beads on a string.

🔑 Key Idea

Humans have **46 chromosomes** (23 pairs) in each body cell. You got **23 from each parent** — which is why you inherit a mix of traits from both.

3.3 Dominant and Recessive

Genes often come in different versions called **alleles**. For each trait, you have two alleles — one from each parent. Some alleles are stronger (**dominant**) and some are weaker (**recessive**).

We use letters to show alleles. A **capital letter** stands for the dominant allele, and a **lowercase letter** stands for the recessive one. For example, with tongue rolling:

- **T** = can roll tongue (dominant)
- **t** = cannot roll tongue (recessive)

Combination	Name	Result
TT	Homozygous dominant	Can roll tongue
Tt	Heterozygous	Can roll tongue (T hides t)
tt	Homozygous recessive	Cannot roll tongue

! Genotype vs. Phenotype

The **genotype** is the gene combo (like Tt). The **phenotype** is the trait you can see (like "can roll tongue"). A recessive trait only shows when **both** alleles are lowercase (tt).

3.4 Punnett Squares

A **Punnett square** is a tool that predicts the chances of offspring inheriting a trait. It shows every possible combination of alleles from two parents.

		Parent 1: Tt	
		T	t
Parent 2: Tt	T	TT	Tt
	t	Tt	tt

Cross: Tt × Tt

Both parents can roll their tongues but each carries one recessive allele. The square shows four equally likely outcomes:

- **TT** — 1 out of 4 (25%) can roll
- **Tt** — 2 out of 4 (50%) can roll
- **tt** — 1 out of 4 (25%) cannot roll

So there's a **75% chance** of a tongue-roller and a **25% chance** of a non-roller.

★ How to Build a Punnett Square

- Write one parent's alleles across the top.
- Write the other parent's alleles down the left side.
- Fill each box by combining the letter from its row and column.
- Count the results to find the probability of each outcome.

3.5 Asexual Reproduction

Asexual reproduction needs only **one** parent. The offspring is a **clone** — it has exactly the same DNA as the parent. There is no mixing of genes.

Type	How It Works	Example
Binary fission	One cell splits into two identical cells	Bacteria
Budding	A small bud grows off the parent, then breaks away	Hydra, yeast
Regeneration	A whole new organism grows from a piece	Starfish, planaria
Vegetative	New plants grow from roots, stems, or leaves	Strawberry runners

 **Pros**

Fast, needs no mate, and works when only one organism is around. Great in stable environments.

 **Cons**

No variety. Because every offspring is identical, one disease could wipe out the whole group.

3.6 Sexual Reproduction

Sexual reproduction needs **two** parents. Each parent gives half of its DNA, so the offspring is a unique mix — not identical to either parent. This creates **genetic variation**.

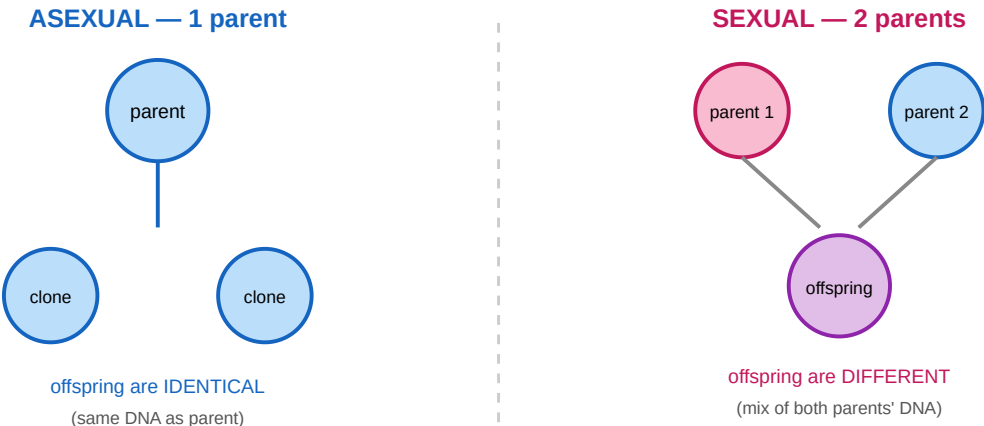


Figure 3.2 — Asexual reproduction makes clones; sexual reproduction makes variety.

Feature	Asexual	Sexual
Number of parents	One	Two
Offspring DNA	Identical (clones)	Unique mix
Genetic variation	None	Lots
Speed	Fast	Slower

 **Why Variation Matters**

Genetic variation is like not putting all your eggs in one basket. If the environment changes or a new disease appears, variety means **some** offspring are likely to survive — and that helps the species continue.

3.7 Selective Breeding

Humans can steer which traits get passed on. In **selective breeding**, people choose which organisms reproduce so that desired traits show up more often in the next generation.

This is how we got most farm animals, pets, and crops. People picked the cows that gave the most milk, the dogs that were friendliest, and the corn with the biggest kernels — and bred those together, again and again, over many generations.

Started With	Selectively Bred Into
Wild wolves	Hundreds of dog breeds, from Chihuahuas to Great Danes
Wild mustard plant	Broccoli, cabbage, kale, and cauliflower
Small wild corn (teosinte)	The large sweet corn we eat today

! Did You Know?

Broccoli, cauliflower, cabbage, and kale all came from the **same** wild plant. Farmers just bred for different parts — flowers, leaves, or stems.

✗ Common Mix-Up

Selective breeding is **not** the same as natural selection. In selective breeding, **humans** choose the traits. In natural selection, the **environment** does.

3.8 Putting It All Together

Traits pass from parents to offspring through genes carried on chromosomes. Punnett squares help us predict which traits offspring might inherit. Asexual reproduction makes identical clones, while sexual reproduction creates variety. And humans use selective breeding to shape traits on purpose.

Big Idea	What to Remember
Heredity	Traits pass from parents to offspring through genes.
Gene structure	DNA → genes → chromosomes, all inside the cell's nucleus.
Alleles	Dominant (capital) hides recessive (lowercase). Recessive shows only as tt.
Punnett squares	Predict the probability of offspring traits.
Asexual vs sexual	One parent = clones; two parents = genetic variation.
Selective breeding	Humans choose which traits get passed on.

★ Standard Check — S7L3

You can now **(a)** explain the role of genes and chromosomes in inheritance, **(b)** model how asexual reproduction makes identical offspring while sexual reproduction makes variation, and **(c)** describe how humans use selective breeding. Standard complete!